

Math Class 9th Federal Board

Chapter 11 Basic Statistics

PAST Papers & Mode Paper

Long Questions

→ CH #4, 6, 8, 9, 11

A school principal wants to allocate funds to different departments based on student enrolment. The number of students in each department is tabulated as under.

Subject	x	Science	Maths	English
No. of Students	f	200	150	180

If the total available fund is Rs 700,000, calculate the amount each department should receive if funds are distributed equally per student.

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Sol: ∴ Total funds = Rs. 700,000

Total Students = $200 + 150 + 180$
 $= 530$

Fund per Student = $\frac{700,000}{530}$

$$\overset{\text{Rs.}}{=} 1320.7547$$

1) Science Department Fund

$$= \overset{\text{Rs.}}{200 \times \frac{1320.7547}{\text{Student}}}$$

$$= \text{Rs. } 264150.9434 \checkmark$$

2) Math Dept. Funds.

$$= 150 \times 1320.7547$$

$$= \text{Rs. } 198113.205$$

3) English Dept. Funds

$$\overset{\text{Rs.}}{=} 180 \times 1320.7547$$

$$= \text{Rs. } 237735.846$$

$$\bar{x} = \frac{\sum fx}{\sum f}$$

Consider the data and find the mean.

✓ CI	140-150	150-160	160-170	170-180	180-190
f	6	9	13	7	5

Grouped

4+4

Sol:-

C-I	f	x Class Mark	fx
140-150	6	145	870
150-160	9	155	1395
160-170	13	165	2145
170-180	7	175	1225
180-190	5	185	925
$\sum f = 40$			$\sum fx = 6560$

$$\text{Mean} = \bar{x} = \frac{\sum fx}{\sum f} = \frac{6560}{40}$$

$$\boxed{\text{Mean} = 164} \quad R_2$$

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The daily consumption of electricity units of 35 factory machines is recorded as in given table. Calculate mode of the data.

C-I	20-24	25-29	30-34	35-39	40-44
f	8	12	2	10	3

Sol: ✓

25, 26, 27, 28, 28 ✓

C-I	f	Class Boundaries	
20-24	8 f_1	19.5	24.5
25-29	12 f_m	24.5	29.5
30-34	2 f_2	29.5	34.5
35-39	10	34.5	39.5
40-44	3	39.5	44.5

⇒ → Modal Class

For grouped data

$$\text{Mode} = l + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times h$$

$$2f_m - f_1 - f_2 \rightarrow \textcircled{1}$$

As 12 is the highest frequency,

∴, modal class = 25-29

$$l = 24.5, \quad f_m = 12$$

$$f_1 = 8, \quad f_2 = 2, \quad h = 5$$

Put the values in Eqn (1), we get

$$\text{Mode} = 24.5 + \left(\frac{12 - 8}{2(12) - 8 - 2} \right) \times 5$$

$$= 24.5 + \left(\frac{4 \times 5}{14} \right)$$

$$= 24.5 + \underline{\underline{\frac{20}{14}}}$$

$$= 24.5 + \frac{20}{44}$$

$$\Rightarrow \boxed{\text{Mode} = 25.93}$$

The given table shows distribution of marks obtained by 50 students in a mathematics test:
Calculate the median marks.

C-I	01-10	11-20	21-30	31-40	41-50
f	5	10	15	12	8

درج جدول میں ریاضی کے ٹیسٹ میں 50 طلبہ کے حاصل کردہ نمبروں کی تفصیل ہے۔ اس مواد کا وسطانیہ معلوم کریں۔

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Computation
f.g.

Sol:

C-I	f	Class Boundaries	Cf
1-10	5	0.5-10.5	5
11-20	10	10.5-20.5	15
21-30	15	20.5-30.5	30
31-40	12	30.5-40.5	42
41-50	8	40.5-50.5	50

For a Grouped Data

For a frequency distribution

$$\text{Median} = l + \frac{h}{f} \left(\frac{N}{2} - c \right) \quad \text{---} \textcircled{1}$$

As $N = 50$, $\frac{N}{2} = 25$

As $\frac{N}{2} \rightarrow$ Mid value lies in
the 21-30 Class - interval i.e.,
21-30 is the Median Class

So, $l = 20.5$, $h = 10$,

$f = 15$, $C = 15$

Put all these values in Eq. (1)

$$\text{Median} = 20.5 + \frac{10}{15}(25-15)$$

$$= 20.5 + 6.67$$

$$\boxed{\text{Median} = 27.167}$$

A box contains 2 red, 4 blue, 6 green and 8 yellow balls. A ball is drawn at random. Find the probability of each color.

sol. Total balls = $2 + 4 + 6 + 8$

$$= 20$$

$$P(\text{Red}) = \frac{2}{20} = \frac{1}{10}$$

$$P(\text{Blue}) = \frac{4}{20} = \frac{1}{5}$$

$$P(\text{Green}) = \frac{6}{20} = \frac{3}{10}$$

$$P(\text{Yellow}) = \frac{8}{20} = \frac{2}{5}$$

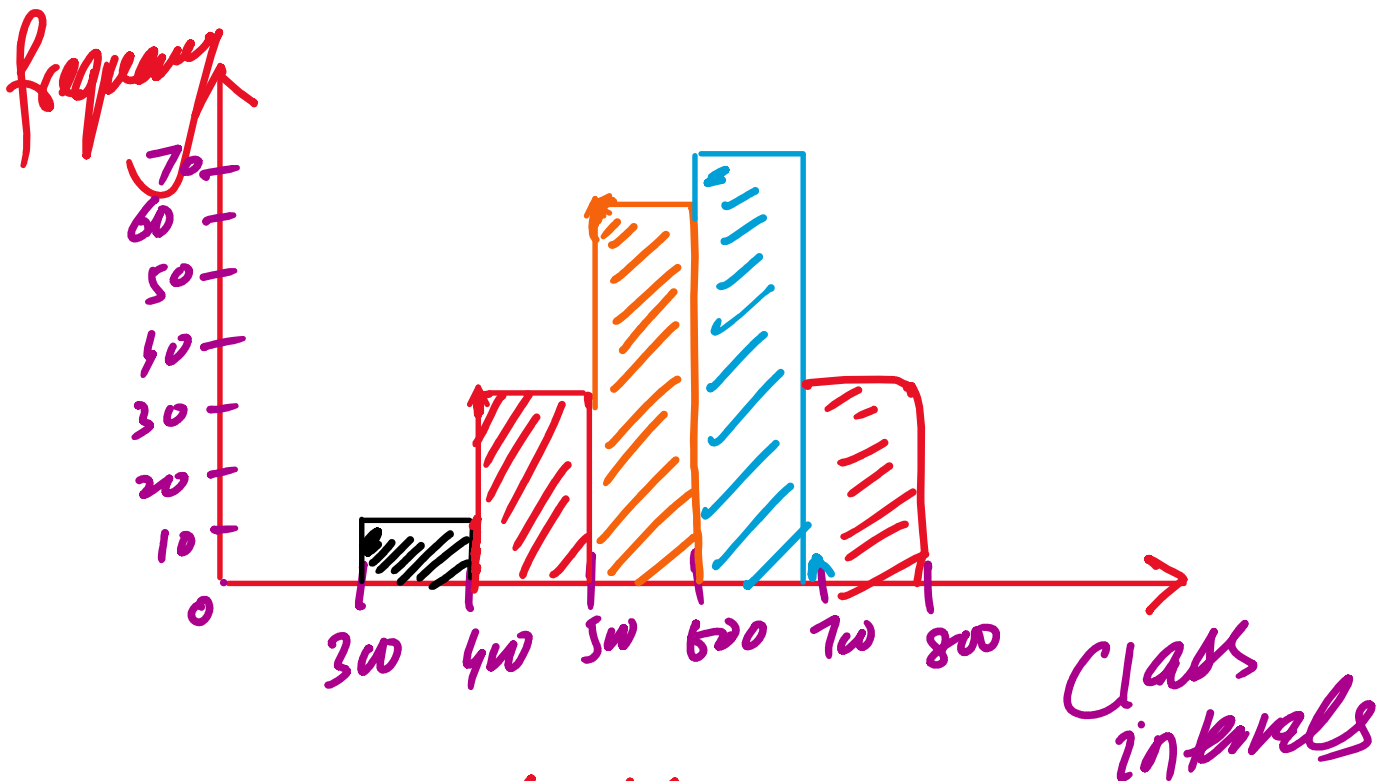
$$P(\text{Yellow}) = \frac{20}{200} = \frac{1}{10}$$

A factory produces 200 bulbs daily. Their lifetimes are grouped in a frequency table. Construct a histogram. Find modal class and modal frequency.

Lifetime (in hours)	Frequency
300 - 400	10
400 - 500	30
500 - 600	60
600 - 700	70
700 - 800	30

Continuous

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Modal Class

600 - 700

600 - 70

Modal Freq. = 70